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Master's Thesis

Sentinel

Automated Self-Belay System

An automated electronic climbing belay device that manages rope slack and performs fast-acting emergency braking, removing the dependency on a continuously attentive human belayer.

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Abstract

Sentinel is a mechatronic system designed to function as an automated electronic climbing belay device. Unlike conventional belay devices, which still require a human operator to feed rope and arrest falls, *Sentinel* aims to autonomously manage rope slack during a climber's ascent and to trigger a fast-acting mechanical brake upon detecting a fall. This thesis documents the design, implementation, and evaluation of a functional prototype, together with an analysis of the regulatory and safety context relevant to automated belay devices.

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Chapter 1

Introduction

The name *Sentinel* is derived from the Old Italian word *sentinella*, referring to a person or thing tasked with watching over, protecting, and alerting others of potential danger. This is precisely what *Sentinel* aims to do for a climber: it continuously watches over the climbing rope, manages it to protect the climber during the ascent, and reacts immediately – by braking – if a fall is detected.

1.1 Background

Climbing is a sport and activity that involves moving upward on rock faces or artificial walls using hands and feet for support, strength, and balance. There are several main disciplines: **bouldering**, which is performed on short walls without ropes; **sport climbing**, where a rope and pre-placed bolts protect the climber from falls; and **traditional (trad) climbing**, in which the climber places their own protection as they ascend. In sport and trad climbing, as well as multi-pitch routes, the rope and safety systems are fundamental for fall protection and security.



Figure 1.1: Sport climber on rope with belayer below.

In these disciplines, a *belayer* manages the rope from below, feeding or taking in slack as required, and holding the rope in case the climber falls. The belayer is essential for safety: they must always pay close attention, anticipate the climber's movements, and react instantly to any sudden load on the rope. Managing slack and braking are continuous, active processes.

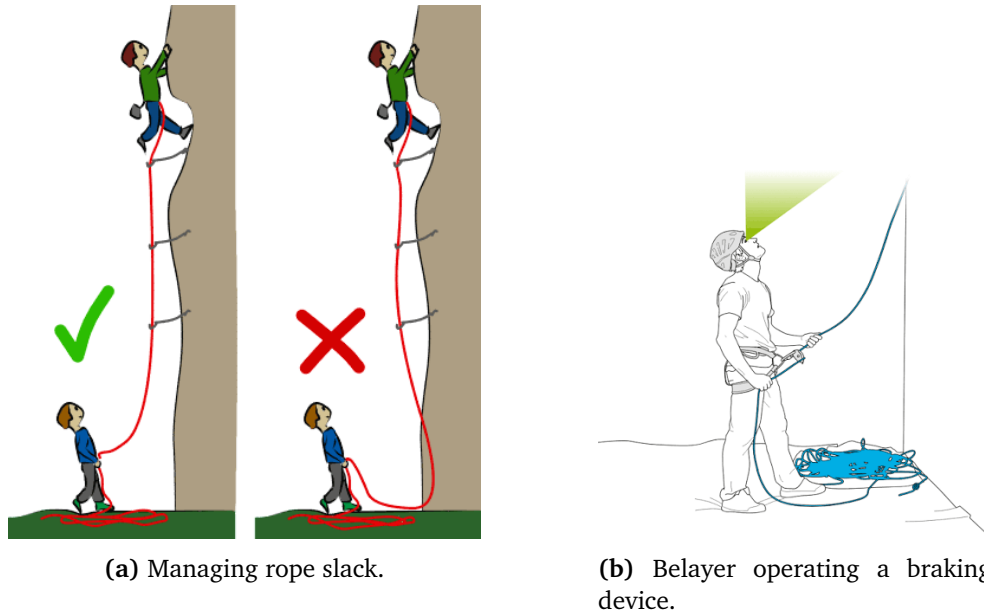


Figure 1.2: The belayer: (a) adjusts rope slack, (b) operates gear to catch a fall.

To connect the rope to the anchor and the belayer's harness, a wide variety of belay devices exists. They differ in how much braking force they provide on their own and in how much continuous attention the belayer must still supply. At the simplest end are **figure-eight descenders**, such as the CAMP Otto [1]: the rope is woven through a fixed aluminium shape and braking relies entirely on rope friction and the belayer's brake-hand grip, with no mechanical assist when a load suddenly appears (Figure 1.3).



Figure 1.3: CAMP Otto figure-eight descender.

A step above are **manual belay devices**—tube-style designs such as the Petzl Reverso [2]. The rope is bent through the device and, when loaded, is pressed into V-shaped friction grooves that increase holding force, but catching a fall still depends completely on the belayer maintaining correct hand position and pulling the

brake strand downward (Figure 1.4). These devices are lightweight and versatile, yet they offer no additional help beyond friction geometry: slack management and the decision to lock the rope remain fully manual.



Figure 1.4: Petzl Reverso manual belay device.

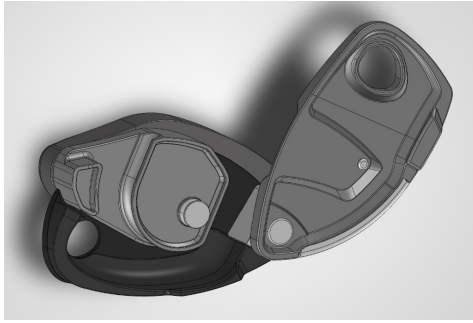
Assisted Braking Devices (ABDs) go further by adding a mechanical response when the rope is loaded quickly. The belayer must still feed and take in slack, but the device itself contributes extra holding force during a fall or hard catch. Within this family, designs differ in whether they use internal moving parts. **Geometric ABDs** such as the Mammut Smart 2.0 [3] have no moving components: under load, the rope is forced into a shaped channel and the carabiner orientation increases friction passively. That geometry assists a skilled belayer but provides no discrete latch or lever that an actuator could command (Figure 1.5).



Figure 1.5: Mammut Smart 2.0 geometric assisted-braking device.

Active ABDs add internal moving parts that react to rope motion or tension. The Petzl GriGri [4] uses a pivoting cam that pinches the rope against a grooved sheave when a sudden load is applied; the Wild Country Revo [5] uses a centrifugal wheel that locks when the rope accelerates rapidly in either direction (Figure 1.6). Both reduce belayer effort during a catch, but their internal layouts differ sharply: the GriGri exposes a single cam lever with a clear open/closed state, whereas the Revo packs an inertial wheel, clutch, and locking elements into a compact, tightly coupled assembly.

For Sentinel, the GriGri-style cam architecture is taken as the mechanical reference. Its pinching mechanism is mechanically straightforward to model, has a



(a) Petzl GriGri (cam-assisted).



(b) Wild Country Revo (inertia-assisted).

Figure 1.6: Active assisted-braking devices with internal moving parts.

natural motor actuation point on the cam lever, and retains a proven passive lock-on-load behaviour that can serve as a fail-safe when power is lost. The Revo's inertial internals would be far harder to motorise and instrument in a first prototype, and the Smart's geometry-dependent braking offers no comparable actuation interface. Automating a GriGri-inspired cam mechanism therefore strikes the best balance between mechanical simplicity, field-proven safety, and suitability for active motorisation.

1.2 Problem Area

However, reliance on a human belayer introduces risks: errors due to fatigue, distraction, or inexperience can result in too much slack or a delayed braking response. Furthermore, climbing requires one person to belay while the other climbs, meaning both cannot climb simultaneously. Automating the belaying process could increase safety, reduce human error, and allow both climbers to be on the wall at once.

This thesis addresses the question: *Can the key safety functions of a human belayer be reliably replaced by an autonomous, mechatronic system?* Such a system would have to actively assess the climber's state, manage rope slack, detect falls, and operate a braking mechanism in real time, all without human intervention. This challenge requires seamless integration of mechanical engineering (for rope handling and braking), sensors (to monitor movement and tension), and robust embedded software to ensure immediate, safe operation and fail-safe behaviour, all while adhering to strict climbing equipment standards.

If successful, such a device could enable new forms of climbing—such as both partners climbing simultaneously—and greatly reduce the risk posed by belayer

error, fatigue, or distraction.

1.3 Narrowing

The problem posed in Section 1.2 applies in principle to every roped climbing discipline, environment, and stage of product development. This thesis does not address that full space. The problem is narrowed to the following boundary conditions:

- **Climbing context:** sport climbing with a single belay from below. Bouldering (no rope) and trad climbing (climber-placed protection) lie outside the problem frame.
- **Operating environment:** bench-top rigs and controlled indoor tests, not outdoor crags or operational deployment in commercial climbing gyms.
- **Rope assumptions:** one standard dynamic rope diameter and type, not the full range of ropes or belay configurations on the market.
- **Solution maturity:** proof-of-concept level—a laboratory demonstrator rather than a manufacturable or certified consumer product.

1.4 Objectives

1.4.1 Principal Objectives

1. Create a functional prototype of an automated self-belaying device.
2. Perform comprehensive testing and evaluation of the prototype.

1.4.2 Secondary Objectives

1. Analyze regulations, standards, and safety requirements applicable to such climbing devices.
2. Survey existing commercial and academic self-belay and assisted-braking solutions.
3. Formulate comprehensive functional and technical requirement specifications for the system.

1.5 Research Questions

- RQ1** To what extent can an embedded sensing system reliably distinguish between a climber pulling in rope during normal ascent and the onset of an actual fall?
- RQ2** How can an automated belay mechanism safely transition out of a braked state once the climber has resumed a valid climbing position, without requiring manual intervention?

- RQ3** Which existing regulations and standards applicable to climbing protection equipment are relevant to an automated belay device, and what safety requirements do they impose on its design?
- RQ4** What mechanical and control-system safeguards are necessary to guarantee a safe fallback behaviour (e.g. automatic rope locking) in the event of a power or electronic failure?

1.6 Limitations

The following are explicitly outside the scope of this thesis, unless revisited as future work:

- Formal certification of the device against climbing safety standards (e.g. UIAA, CE/EN).
- Outdoor field testing with real climbers at height.
- Support for multiple rope diameters, types, or conditions (e.g. wet or icy rope).
- Weatherproofing and long-term durability testing of the prototype enclosure.
- Mass-manufacturability and industrial design for a commercial product.

1.7 Contributions

This thesis contributes the following:

1. A working proof-of-concept prototype that demonstrated automated, sensor-driven belaying with real-time slack management and fast emergency braking is technically feasible using accessible embedded hardware.
2. An analysis of existing commercial and academic self-belay solutions, situating this project's approach relative to the state of the art.
3. A documented functional and technical requirement specification for automated belay devices.
4. An analysis of applicable regulations, standards, and safety requirements relevant to automated belay device design.
5. Empirical performance data (e.g. reaction time, braking distance, false-trigger rate) characterising the prototype's safety-relevant behaviour under controlled test conditions.

1.8 Report Structure

This report is organised to build from problem definition to implementation and evaluation. The chapters are structured as follows:

- Chapter 1 introduces the motivation, problem definition, narrowing, objectives, research questions, and scope.

- Chapter 2 defines the functional and non-functional requirements that the system must satisfy, at both a high and low level.
- Chapter 3 describes the overall architecture of the chosen solution and the rationale behind it.
- Chapter 4 documents the development process followed, including tools for project management and documentation.
- Chapter 5 covers the technical details of the implementation: tools, languages, libraries, and representative code or design examples.
- Chapter 6 describes how the system is set up and deployed on the target hardware.
- Chapter 7 describes how the system was tested and what was learned from testing.
- Chapter 8 discusses the process and product choices made throughout the project.
- Chapter 9 summarises what was achieved and outlines future work.

1.9 Report Tools

This section documents the tools used in preparing this report and the underlying project work.

1.9.1 AI Tools

An AI-assisted coding and documentation environment (Cursor) was used throughout the project to accelerate firmware scaffolding, documentation drafting, and LaTeX authoring. All AI-assisted contributions were reviewed and directed by the author; ambiguous design decisions were resolved by the author rather than left to automated assumptions, per the project's development workflow.

1.9.2 CAD Tools

Mechanical design and reference modelling work uses SolidWorks, alongside publicly available reference CAD models (see Section 1.1) used for early-stage design inspiration.

1.9.3 LaTeX and Documentation Tools

This report is written in $\text{\LaTeX}$, using the community `ntnuthesis` document class for the long-form thesis, compiled locally with a MiKTeX distribution (`pdflatex/biber`). General project documentation (hardware specifications, pinouts, and development history) is maintained in Markdown under `docs/`.

1.9.4 Software Development Environments

Embedded firmware is developed in C++ using the PlatformIO build system and toolchain, targeting an embedded microcontroller platform. Version control is managed with Git.

Chapter 2

Functional Requirements

This chapter defines the functional requirements that the Sentinel system must satisfy. Requirements are described functionally, independent of the specific hardware components used to realise them; component selection and justification are deferred to dedicated design chapters later in this report.

2.1 Climbing Mechanics and Slack Management

The system must automatically take in rope as the climber ascends, so that the amount of slack in the system remains within a safe, comfortable range without manual intervention. To do so, the system must reliably distinguish between two dynamically different situations that both involve rope moving through the mechanism:

- **Normal ascent / rope pull-in**, where the climber is moving upward and the rope must be taken in at a corresponding, comparatively gradual rate. The device also has to be able to allow rope taking during climbing without blocking the system.
- **A fall event**, characterised by a sudden increase in rope acceleration and a rapid increase in rope tension, which must be distinguished from normal climbing dynamics (e.g. clipping, resting on the rope, or bursts of quick movement).

This distinction must be made using onboard rope-movement and tension-sensing information, processed in real time, and must be robust enough to avoid both missed fall detections and false-positive brake triggers during ordinary climbing.

2.2 Post-Fall State Detection

After the system has arrested a fall by engaging the brake, it must be able to detect when the climber has resumed a valid, stable climbing position (e.g. has regained

a hold and is no longer hanging dynamically on the rope). Once this state is detected, the system must release the brake and resume normal slack-management behaviour automatically, without requiring a manual reset from the climber or a belay partner, unless a manual reset is explicitly designed as a deliberate safety fallback. Also this has to be able to do by pressing a button in the remote control.

2.3 Operational Modes and Remote Control

The system must support at least two distinct operational modes:

- **Climbing Mode:** the default mode described in Section 2.1 and Section 2.2, in which the system manages slack during ascent and arrests falls.
- **Rappel Mode:** a mode that allows a controlled, continuous descent, with the braking behaviour adapted for a deliberate downward motion rather than treating it as a fall event.

Switching between these modes must be possible via a wireless remote control, allowing the operator (climber or belay supervisor) to change modes without physically accessing the device mounted on the anchor point.

The rappel will be done by maintaining a downwards button in the remote controll. There must be an other button in the remote controll to pull the rope.

2.4 Battery Management and Safety Protocols

2.4.1 Battery Level Indication

The system must visually and clearly indicate the current battery charge level at all times during operation, so that the user can make an informed decision about whether it is safe to begin or continue climbing.

2.4.2 Low-Battery Restrictions

The system must enforce the following safety protocols based on battery state:

- If the battery level is low at start-up, the system must prevent Climbing Mode from being initiated.
- If the battery level drops to a low or critical level *during* an active climbing session, the system must restrict its functionality strictly to Rappel/Abseil Mode, allowing the climber currently on the wall to be safely lowered rather than left in an unsupported state. The system must do a clear sonoric advice to announce the mode switch.

2.5 Fail-Safe Mechanical Override

This requirement is considered **critical** to the safety case of the system.

2.5.1 Automatic Rope Lock on Power Loss

In the event of a power loss or an electronic/software failure, the rope must lock automatically by default. The mechanical design must guarantee this fail-safe behaviour independent of the control software – that is, the default, unpowered, or faulted state of the mechanism must be “locked”, not “free-running”.

2.5.2 Manual Override Lever

A manual physical lever must allow the climber (or a belay supervisor) to override the automatic lock and execute a controlled descent by hand, even with no power available. This requires evaluating a passive, GriGri-like mechanical behaviour that functions correctly without any electronic assistance, ensuring that a power or electronics failure never leaves a climber stranded with no means of controlled descent.

2.6 Non-Functional Requirements

Beyond the functional behaviour described above, the system must also satisfy a number of non-functional requirements. These are not yet exhaustive and are expected to be expanded as the design matures; initial candidates include:

- **Reliability and safety-critical performance:** given that a missed or delayed fall detection has direct safety consequences, the system’s sensing and braking response must meet a bounded, worst-case reaction latency, and must degrade towards the fail-safe locked state (Section 2.5) rather than towards an unsafe state under any fault condition.
- **Real-time responsiveness:** sensor acquisition, state estimation, and brake actuation must run within firm real-time deadlines, since the usefulness of the system depends on reacting fast enough to arrest a fall before excessive slack develops.
- **Usability and ergonomics:** the device must be operable by a climber with no more training than is required for a conventional assisted-braking belay device, with clear, unambiguous feedback (e.g. battery level, operating mode) that does not require reading a manual mid-climb.

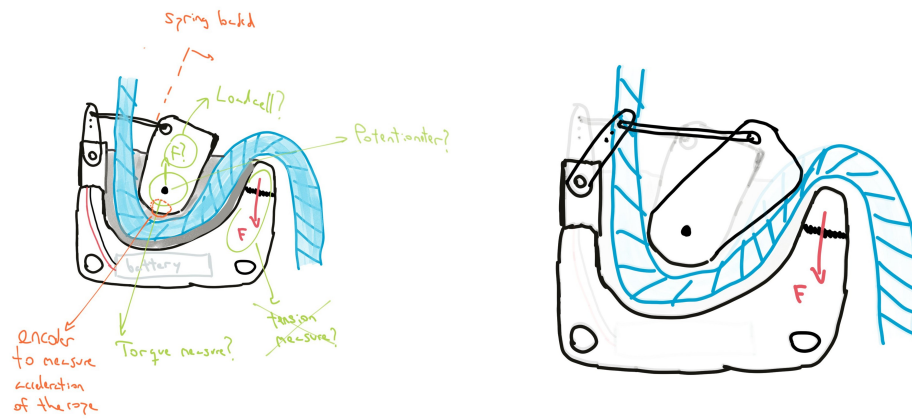
Chapter 3

Technical Design

3.1 Conceptual Design Sketch (Phase 1)

Figure 3.1 shows early-stage hand sketches produced during Phase 1 of the mechanical design, used to visualise which functions the rope-pinching mechanism needs to provide before committing to a detailed CAD model. Figure 3.1a marks the required components; Figure 3.1b shows the same mechanism in an actuated state. The sketches identify the following elements:

- An **encoder**, used to measure the rope's acceleration as it moves through the mechanism.
- A **load pin**, integrated into the pivot axle of the pinching lever, used to measure the tension on the climbing side of the rope (i.e. the side leading up to the climber). This tension reading provides the information needed to determine when the climber has regained a stable position on the wall, so that the pinch can be safely released (see Section 2.2).
- A **spring** built into the pinching mechanism, providing the passive fail-safe behaviour described in Section 2.5: the mechanism defaults to a locked/pinched state on its own, without requiring power.
- A **servo**, used to actuate the pinching lever – i.e. to open and close the brake under motor control.
- **Axles for the outer cover**, and a **hole for the carabiner** used to attach the device to the anchor point.
- An **accommodation volume for the electronics**, i.e. the internal space reserved for the control board and wiring.
- A **potentiometer**, mounted in the pinching mechanism, used to sense the lever's current angular position (i.e. how open or closed the brake currently is).



(a) Device with the required functions marked.

(b) Device when the pinching mechanism is actuating.

Figure 3.1: Phase 1 conceptual sketches of the rope-pinching mechanism.

Chapter 4

Development Process

Chapter 5

Implementation

Chapter 6

Deployment

Chapter 7

Testing and User Feedback

This chapter documents bench-top tests used to evaluate the performance of the Sentinel prototype.

7.1 Testing of traditional assisted braking devices

To be legally commercialized as Personal Protective Equipment (PPE) within the European Union and recognized globally, assisted braking belay devices must comply with rigorous safety standards and standardized testing protocols. Because these devices form a critical link in the climbing safety chain, they are classified under Category III PPE according to Regulation (EU) 2016/425 [eu_ppe_2016], governing equipment designed to protect users against risks that may cause severe consequences such as death or irreversible health damage.

7.1.1 Applicable Standards and Certifications

Commercial assisted braking devices (ABDs) must satisfy three primary normative frameworks prior to market entry:

- **Regulation (EU) 2016/425 (CE Marking):** The legal framework for PPE distributed in Europe. Category III compliance mandates stringent quality control, including independent type-examination by a Notified Body (Module B) and ongoing manufacturing quality monitoring (Module C2 or D) [eu_ppe_2016].
- **EN 15151-1 (Type 8 / Type 5):** The specific European standard defining safety requirements and laboratory testing procedures for braking devices with manually assisted locking [en15151_1].
- **UIAA 129 Standard:** The safety standard published by the International Climbing and Mountaineering Federation (*Union Internationale des Associations d'Alpinisme*). The UIAA standard builds upon EN 15151-1 but mandates static structural load thresholds between 15 kN and 22 kN to ensure an adequate safety margin under extreme dynamic loading scenarios

[uiaa129, TFM_Sentinel_2026].

7.1.2 Standardized Testing Procedures

Under standard EN 15151-1, prototype and production units are subjected to strict mechanical and functional assays in laboratory settings:

- 1. **Dynamic Drop Test (Impact Performance):** A standardized rigid mass of 80 kg is dropped in a test rig simulating a severe lead fall. Across three consecutive test drops, the mean rope slippage through the device locking mechanism must not exceed 1.5 m before fully arresting the load [en15151_1].
- 2. **Static Holding Force Test:** With the locking mechanism engaged, a static tensile force of 8 kN is applied to the device for 60 seconds. The unit must sustain this load without structural rupture or releasing the rope. In actual climbing falls, dynamic impact forces transmitted to the belay device typically reach between 4 kN and 8 kN [TFM_Sentinel_2026].
- 3. **Locking Engagement Test:** A lower static load of 2 kN is applied for 60 seconds to verify that the assisted braking mechanism triggers reliably under moderate rope tension. The maximum allowable rope displacement during engagement is 300 mm.
- 4. **Anti-Panic Mechanism Test (Type 8 Devices):** For devices equipped with an anti-panic lowering lever, tests verify that pulling the lever completely past its operational position—a panic reflex scenario—automatically re-engages the braking mechanism.

7.1.3 Summary of Compliance Thresholds

Table 7.1 outlines the required physical tests, key test parameters, and mandatory compliance criteria required by EN 15151-1 and UIAA 129.

Table 7.1: Standardized safety tests and pass criteria for assisted braking devices.

Test Type	Test Parameters	Mandatory Pass Criteria
Dynamic Performance	80 kg drop mass, 3 iterations	Max rope slippage ≤ 1.5 m
Static Holding Force	8 kN tension for 60 s	No structural failure or rope release
Locking Activation	2 kN load for 60 s	Activation with slippage ≤ 300 mm
Ultimate Tensile Load	Pull to structural failure	Must withstand $\geq 15\text{--}22$ kN [uiaa129, TFM_Se
Anti-Panic Function	Full lever pull past limit	Automatic brake re-engagement

7.2 Test 1: Weight Hanging

7.2.1 Introduction

The first test checks the most basic safety function of the device: when a load is suddenly applied to the climbing side of the rope, Sentinel must detect the event

and lock the rope before the load travels an unsafe distance. A controlled weight-drop on a bench-top rig provides a simple, repeatable way to verify that the sensing and braking chain responds to a fall-like load without requiring a human climber or a full wall setup.

7.2.2 Methods

The Sentinel prototype is mounted on a fixed anchor point with a dynamic climbing rope threaded through the mechanism. A known mass is attached to the climber side of the rope and held at a defined height so that the rope carries little or no tension while the device is in its normal operating state. The mass is then released, applying a sudden load to the rope.

For each trial, the following is recorded:

- Whether Sentinel engaged the brake and arrested the load.
- The time from release to brake engagement (reaction time), measured from the sensor/log output or a high-speed recording of the test.

The test is repeated for several trials to check basic repeatability. This first version establishes the procedure; thresholds, mass values, and drop height will be refined in later iterations.

7.2.3 Materials

- Sentinel prototype (firmware flashed and powered).
- Standard dynamic climbing rope (diameter as specified in Section 1.3).
- Fixed anchor structure or test frame.
- Hanging mass of known weight (e.g. kettlebell or plate).
- Means of holding and releasing the mass without imparting extra impulse (e.g. quick-release hook or manual cut-away).
- Timing/logging equipment (serial log timestamp, oscilloscope trigger, or video at known frame rate).

7.2.4 Results

7.2.5 Discussion

Chapter 8

Discussion

Chapter 9

Conclusion

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